

YOLOv11 Environment Construction

YOLOv11 Environment Construction

1. Overview
2. Environmental requirements
3. Docker setup
 - 3.1. Official installation of Docker
 - 3.2. Alibaba Cloud install Docker
 - 3.3. Add access rights
4. Install Ultralytics
5. Edit the startup script
6. Execute the script
- References

This chapter is applicable to self-construction of Jetson nano official image. If you use the YAHBOOM version of the image, this tutorial can be ignored.

1. Overview

YOLO11 is Ultralytics YOLO is the latest product in the real-time object detector series, redefining what is possible with state-of-the-art accuracy, speed, and efficiency. Building on the impressive progress of previous YOLO versions, YOLO11 has made significant improvements in architecture and training methods, making it a versatile choice for a wide range of computer vision tasks.

2. Environmental requirements

Due to limitations, only the older version of **JetPack**4.6.6 can be burned on **jetson nano**. The version used in our image is **JetPack**4.6.3. In this version, **CUDA** 10.2, **cuDNN** 8.2.1, **TensorRT** 8.2.1, and the system comes with **python** version 3.6.

However, the **python** version required to install Ultralytics is 3.8, so this article chooses to use **docker** to install.

3. Docker setup

3.1. Official installation of Docker

If Docker is not installed, you can use the script to install Docker in one click.

Domestic users may not be able to install Docker through official methods. We recommend using Alibaba Cloud to install Docker or directly using our image.

- Update the local package list

```
sudo apt update
```

- Upgrade the installed package

```
sudo apt upgrade
```

- Download and run the script

Download the get-docker.sh file and save it in the current directory.

```
sudo apt install curl
curl -fsSL https://get.docker.com -o get-docker.sh
```

Use sudo privileges to run the get-docker.sh script file.

```
sudo sh get-docker.sh
```

3.2, Alibaba Cloud install Docker

If you cannot install it yourself, please use the image we provide.

- Update the local package list

```
sudo apt update
```

- Install the required software

```
sudo apt install apt-transport-https ca-certificates curl gnupg2 lsb-release
software-properties-common
```

- Add the GPG key of the software source

```
curl -fsSL https://mirrors.aliyun.com/docker-ce/linux/ubuntu/gpg | sudo gpg --
dearmor -o /usr/share/keyrings/docker-archive-keyring.gpg
```

- Add Alibaba Cloud mirror software source

```
echo "deb [arch=arm64 signed-by=/usr/share/keyrings/docker-archive-keyring.gpg]
https://mirrors.aliyun.com/docker-ce/linux/ubuntu bionic stable" | sudo tee
/etc/apt/sources.list.d/docker.list > /dev/null
```

- Install Docker

```
sudo apt update
sudo apt install docker-ce docker-ce-cli containerd.io docker-compose-plugin
```

3.3. Add access rights

Add the current user's access rights to the Docker daemon: You can use Docker commands without using the sudo command

```
sudo usermod -aG docker $USER
newgrp docker
```

4. Install Ultralytics

```
docker pull yahboomtechnology/ultralytics:1.0.3
```

```
jetson@yahboom: ~  
jetson@yahboom:~$ docker pull yahboomtechnology/ultralytics:1.0.3  
1.0.3: Pulling from yahboomtechnology/ultralytics  
f46992f278c2: Pulling fs layer  
d0ec296fcb76: Pull complete  
9e18ddc8ca7a: Pull complete  
457ba495c8e5: Pull complete  
71bca45e35bd: Pull complete  
644761cdc735: Pull complete  
11628dbc31eb: Pull complete  
d364c3700c33: Pull complete  
01869d070b2e: Pull complete  
cc3009375042: Pull complete  
3e182d6364dc: Pull complete  
f72feb4812f9: Pull complete  
151eb940bbec: Pull complete  
0e9dda2495b9: Pull complete  
0e78bdc2f297: Pull complete  
8dc68d594a4e: Pull complete  
777499412cc5: Pull complete  
1ca27931ddcb: Pull complete  
4fb3d314e9f2: Pull complete  
841fe4d09b1d: Pull complete  
bfebf8915158a: Pull complete  
30534ba90a81: Pull complete  
2ed11f6e9536: Pull complete  
33da07fcf1ef: Pull complete  
59ca9260559f: Pull complete  
59ca9260559f: Pull complete  
bd407602c90e: Pull complete  
6bc4831114e2: Pull complete  
c2bb3df72f04: Pull complete  
422737162598: Pull complete  
ab2317f72fc8: Pull complete  
52d86c1d9761: Pull complete  
f356f9e80d33: Pull complete  
5985bf3da46b: Pull complete  
e874e18edb1a: Pull complete  
0a94dac866f8: Pull complete  
10fe5f9b797c: Pull complete  
22bb6505219e: Pull complete  
4e4a9f003c44: Pull complete  
49246b2102eb: Pull complete  
d769d9bc3781: Pull complete
```

Query docker image

```
docker images
```

```
jetson@yahboom: ~  
jetson@yahboom:~$ docker images  
REPOSITORY          TAG          IMAGE ID          CREATED           SIZE  
yahboomtechnology/ultralytics 1.0.3       16d9e8762ca1     24 minutes ago   6.18GB  
yahboomtechnology/ros2-base   2.0.2       bb7c486548cd     18 months ago    5.61GB  
hello-world           latest      b038788ddb22     23 months ago    9.14kB  
jetson@yahboom:~$
```

5. Edit the startup script

[yolov11_docker.sh] The contents of the script are as follows:

```
#!/bin/bash  
xhost +  
  
docker run -it \  
--net=host \  
--env="DISPLAY" \  
--env="QT_X11_NO_MITSHM=1" \  
-v /tmp/.X11-unix:/tmp/.X11-unix \  
-v /home/jetson/temp:/ultralytics/ultralytics/temp \  
--device=/dev/video0 \  
-p 9090:9090 \  
-p 8888:8888 \  
yahboomtechnology/ultralytics:1.0.3 /bin/bash
```

Commented script description:

Note: When adding a host device to the container below, if the host is not connected to the device, you need to remove the corresponding addition operation to start the container

```
#!/bin/bash
xhost +

docker run -it \
--net=host \
--env="DISPLAY" \
--env="QT_X11_NO_MITSHM=1" \
-v /tmp/.X11-unix:/tmp/.X11-unix \
-v /home/jetson/temp:/ultralitics/ultralitics/temp \
--device=/dev/video0 \ # USB camera, add a host device to the container. If the
camera is not connected, please remove this line. Video* needs to be modified
according to the specific device port found in your system
-p 9090:9090 \
-p 8888:8888 \
yahboomtechnology/ultralitics:1.0.3 /bin/bash
```

6. Execute the script

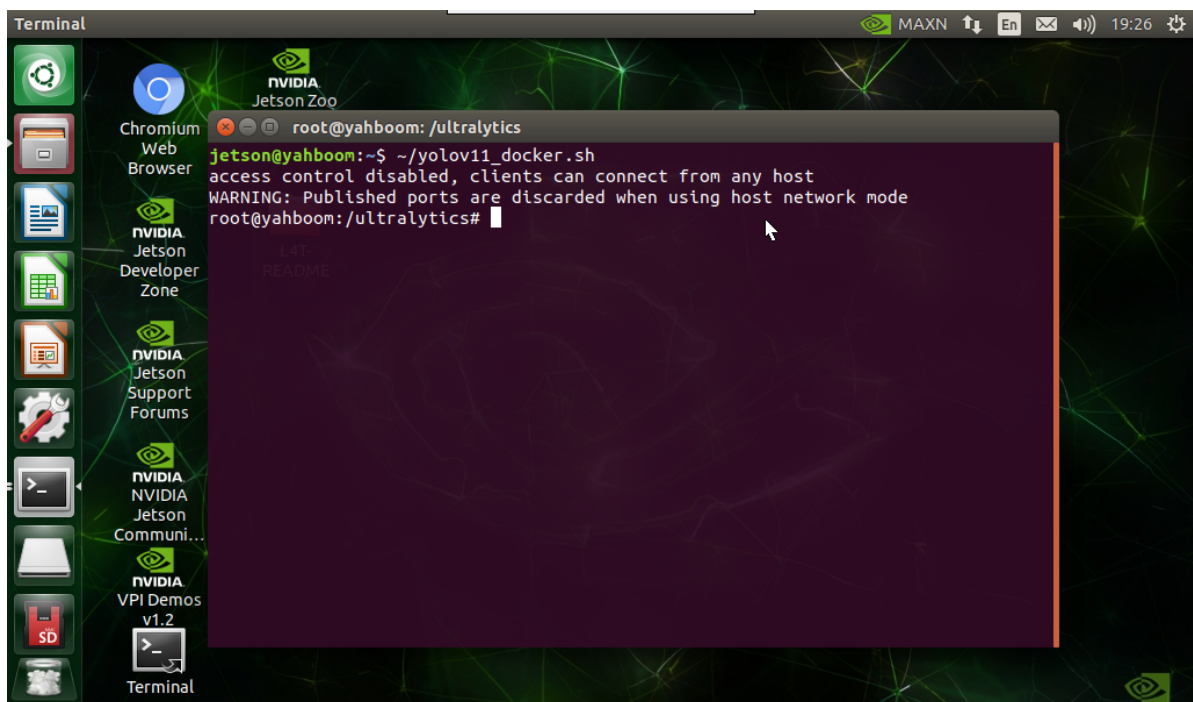
After editing the docker startup script, open the terminal on the docker host (on VNC or on the motherboard display)

Note: This must be executed on VNC or on the motherboard display, and cannot be executed in a terminal accessed remotely via ssh (such as a terminal accessed via MobaXterm), otherwise the GUI image may not be displayed in the container.

Run the previously created startup script in the VNC interface or on the screen (Note: Each execution of the script creates a new container from the image)

```
./yolov11_docker.sh
```

You can enter the container correctly, and run the real-time detection case to display the camera image.



References

<https://docs.ultralitics.com/guides/nvidia-jetson/>

